

Algebra 1
Unit 4
Lesson 6 Practice

Name **Answer Key** _____
Date _____
Period _____

For each real-world representation of a context write an inequality that represents the constraint.

1. A person has to be at least 18 years old to vote.
 $p \geq 18$
2. A job offers a salary between \$42,000 and \$50,000.
 $42,000 < x < 50,000$
3. The highest altitude for skydiving without oxygen is 14,000 feet.
 $s \leq 14,000$

An automatic soap dispenser holds 18 ounces (oz) of soap. Each time someone places their hand under the soap dispenser releases 0.15 oz of soap. The function $f(x) = 18 - 0.15x$ models the amount of soap, in ounces, in the dispenser for x dispenses.

4. Find the value of x for which $f(x) = 13.2$.
 $13.2 = 18 - 0.15x$
 $-18 \quad -18$
 $\frac{-4.8}{-0.15} = -\frac{0.15x}{-0.15}$
 $32 = x$
5. In terms of the context, what does the value of x that is the solution to $f(x) = 13.2$ mean?
 $x(32)$ represents the number of dispenses of soap that are made. So, after 32 dispenses, there are 13.2 ounces remaining.
6. Find the x -intercept.
 $f(x) = 18 - 0.15x$
 $0 = 18 - 0.15x$
 $-18 \quad -18$
 $-\frac{18}{-0.15} = -\frac{0.15x}{-0.15}$
 $x = 120$
7. What does the x -intercept mean within the context?
This means it will take 120 dispenses of soap to have no (or zero) soap left in the dispenser.

8. What key feature is $f(0)$?

This is the y -intercept of the graph. This is the amount of soap when no dispenses are made.

9. Select all of the constraints for the context.

- ☒ $x \geq 0$
- ☐ $x \leq 18$
- ☒ $x \leq 120$
- ☒ $f(x) \geq 0$
- ☒ $f(x) \leq 18$
- ☐ $f(x) \leq 120$

10. Graph the function in terms of the context.

